

# PulseCX AI

SOLO PROJECT

AI-POWERED FEEDBACK ANALYSIS

## Team Composition

**Bhanu Charan** — Solo developer, responsible for workflow automation, AI integration, UI development, and system design.

## Problem Statement

Small teams and student startups receive customer feedback but struggle to extract meaningful insights from it. Without dedicated analytics resources, they are forced to manually read responses, making it difficult to identify recurring issues and take timely action.



# Solution Overview

**PulseCX AI** is an AI-powered agent designed for student startups and small businesses to simplify customer feedback analysis. It automatically collects feedback, processes responses, and transforms them into clear, actionable insights — without requiring a data team.

## Collect

Automatically gathers customer feedback via email campaigns and standardised forms.

## Analyse

Identifies key themes, detects common issues, and performs sentiment analysis using LLM.

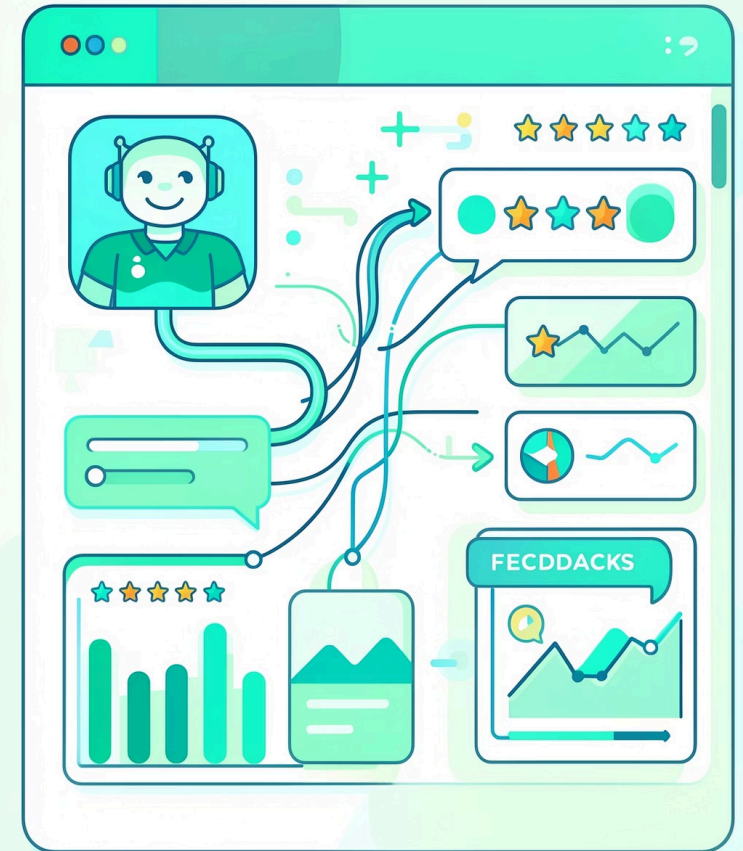
## Prioritise

Generates structured, prioritised recommendations so teams know what matters most.

## Act

Delivers instant insights to the UI, enabling fast, informed decision-making.

 PulseCX AI follows a **modular workflow-based architecture** built to handle feedback collection, processing, and insight generation end-to-end.



# Agent Architecture

PulseCX AI operates across three distinct, interconnected workflows — each handling a critical stage of the feedback pipeline.

1

## Campaign Trigger — Feedback Collection

The system is triggered through a webhook from the user interface. Customer email campaigns are initiated with a feedback form link. Responses are collected through a standardised form and stored in a connected data source.

2

## Feedback Processing & Analysis

A scheduled trigger periodically checks for new, unprocessed feedback entries. Relevant data is filtered and aggregated, then passed to a large language model for sentiment analysis, theme extraction, and issue detection. The system generates structured insights and recommendations. If critical issues are detected, additional analysis is performed. Processed entries are marked as completed to avoid duplication.

3

## Insight Delivery — UI + API

The user interface triggers a webhook request to fetch insights. The system retrieves and aggregates processed feedback data. The language model generates a structured report, and the insights are returned instantly to the UI for display.

Each workflow is independently triggered — via webhooks or scheduled timers — ensuring the system remains responsive, efficient, and free from duplication.

# System Architecture Diagram

The diagram below illustrates the full end-to-end flow of PulseCX AI — from initial setup and campaign dispatch through to AI-powered analysis and instant insight delivery, with Google Sheets serving as the central data store.

## **Real-Time Alerts**

Critical issues trigger instant email alerts to the manager.

## **Automated Insights**

AI-powered reports generated on demand or on schedule.

## **Central Storage**

All feedback responses stored and managed in Google Sheets.

## **No Duplication**

Processed entries are flagged to ensure clean, efficient analysis.

# Decision Logic & Tech Stack

## Decision Logic

### Threshold Check

A minimum feedback threshold ensures insights are generated only when sufficient data is available — preventing premature or misleading conclusions.

### Sentiment Priority

Conditional checks identify negative or critical patterns to prioritise focused recommendations, ensuring the most urgent issues surface first.

### Efficiency

Only unprocessed entries are analysed on each run, preventing duplication and keeping the system lean and accurate.

## Tech Stack



### Workflow Automation

n8n



### Frontend (UI)

HTML, CSS, JavaScript



### AI Model & Inference

Groq API (LLaMA 3.3 70B)



### Data Collection

Google Forms



### Data Storage & Processing

Google Sheets



### Integration Layer

Webhooks (API-based communication)

# PulseCX AI Agent

AI-Powered Customer Feedback Analysis & Action Agent

